# Soham Jagani

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## EDUCATION

**North Carolina State University (NC State), Raleigh, NC** **Expected Graduation: May 2027**
B.S. Mechanical Engineering | Minor in Business Administration

## CERTIFICATIONS

**NCEES:** FE Mechanical

## SKILLS

**Software:** SolidWorks, Ansys, MATLAB, Simulink, AutoCAD, Revit, Inventor, Creo, Maximo

## RELATED EXPERIENCE

**Project Management Intern, Amazon, St. Louis, Missouri** **May 2026 – July 2026**

- Led a zone redesign project to improve operational efficiency and package throughput, developing an AutoCAD layout that increased processing capacity while maintaining compliance with safety standards and generating up to $550,000 in annual cost savings.
- Conducted a comprehensive cost-benefit and ROI analysis for the zone redesign, demonstrating a projected 1-month payback period and validating the financial impact of the proposed improvements.
- Managed the project from initial process evaluation through design and implementation, cross functional collaboration to optimize workflow, improve safety, and increase operational efficiency.
- Monitored real time operational metrics to identify workflow bottlenecks and optimize labor allocation for a team of approximately 30 associates, improving productivity and maintaining safety standards in a high-volume fulfillment environment.

**Manufacturing Engineering Intern, GAF, Arkadelphia, AK** **May 2025 – December 2025**

- Designed a fiberglass dust vacuum system to reduce fiber buildup on equipment, mitigating contamination related wear and extending equipment service life; created detailed SolidWorks technical drawings using GD&T standards to support fabrication and installation.
- Creating and implementing PMs for previously neglected systems, collectively preventing over 100 hours of downtime and saving over $100000 yearly.
- Led inventory updates during a plant-wide Kaizen event, overhauling and standardizing maintenance parts data in Maximo CMMS, while engineering a systematic program to track over 250 rollers and their maintenance requirements.
- Applied Lean Six Sigma principles and analyzed downtime data to identify high-frequency machine failures and recurring issues, developing preventive maintenance procedures that reduced equipment downtime and improved reliability.

**Design Engineer, Soft Body Robotics Research, NC State, Raleigh, NC** **August 2024 – May 2025**

- Utilized SolidWorks and laser cutting technology to create and prototype precision gripper designs.
- Rapidly prototyped and optimized gripper force and structure for reliable stem handling.
- Conducted innovative research on kirigami-based soft robotics for agricultural applications.

**Build Team Engineer, Aerial Robotics Club, NC State, Raleigh, NC** **September 2022 – May 2024**

- Prototyped carbon fiber and Kevlar composite for aircraft wings, optimizing strength-to-weight ratios.
- Create detailed manufacturing drawings, bills of materials (BOMs), and engineering documentation.
- Spearheaded the design and construction of a functional prototype aircraft wing utilizing carbon fiber and Kevlar composites.

## Projects

**PID Control of a Thruster-Actuated Inverted Pendulum** **January 2026 – May 2026**

- Designed Modeled and controlled a nonlinear inverted pendulum/thruster system using MATLAB and Simulink. Developed PID-based simulations for stabilization and disturbance rejection.
- Conducted experimental testing and validation of a pendulum control system by collecting thruster response data, comparing real-world behavior to simulation results, tuning controller parameters, and evaluating system stability under external disturbances.